

Sipna College of Engineering and Technology, Amravati

Branch: - Computer Science & Engineering

Semester: -6th

Subject: - Software Engineering

UNIT – I

- 1) Explain software engineering: a layered technology.
- 2) What are customer's myths? Explain realities of each myth.
- 3) Explain any two software myth related to management myth with its reality.
- 4) What are the four P's of software project management? Explain it in detail.
- 5) What do you mean by S/w process? explain common process framework
- 6) Explain Prototyping model in detail with advantages & disadvantages.
- 7) Explain RAD model in detail with advantages & disadvantages.
- 8) Explain Spiral model in detail with advantages & disadvantages.
- 9) Explain linear sequential model/waterfall model in detail with advantages & disadvantages.
- 10) Explain W5HH principle in detail & also explain three generic team organizations.
- 11) What do you mean by "Software myths"? Discuss in detail.
- 12) What do you mean by evaluation of software? What are the characteristics of software?
- 13) What are practitioner's myths? Explain realities of each myth.
- 14) What are the different factors in choosing software process model?
- 15) Explain umbrella activities in detail.
- 16) What are the attributes of good software?

UNIT-II

- 1) How do you measure software resources?
- 2) Explain the different approaches to sizing problem?
- 3) Explain RIS (Risk Information Sheet) with example.
- 4) Differentiate between process indicators and project indicators
- 5) What are the activities involved in software project planning? Explain each in detail?
- 6) Define measure, measurement and metrics. What are process and project metrics?
- 7) Explain function oriented metric and how, function point is calculated? Give an example?
- 8) Describe differences between risk component & risk drivers.
- 9) What are various software risk explain in detail.
- 10) State and explain the measures of Software Quality.
- 11) List the different metrics for software measurement and explain any one of them.
- 12) Explain the size oriented metrics. Give an example
- 13) Explain how Defect Removal Efficiency for a software product and project is calculated.
Give an example
- 14) Describe COCOMO II Model in detail.
- 15) Describe Risk projection in detail.
- 16) Explain Risk mitigation, monitoring and management.
- 17) Elaborate the concept of Make/Buy decision.

UNIT – III

- 1) Explain various software quality assurance activity in detail
- 2) Write a short note on project scheduling
- 3) Explain with suitable example Earned Value Analysis
- 4) What is Quality Assurance Program Checklist
- 5) State the Project Scheduling principle, mention and elaborate relationship between people and effort.
- 6) What are the minimal set of guidelines for formal technical review
- 7) Explain the empirical relationship between no. of lines of code delivered, effort and development time.
- 8) Explain with example, how timeline chart helps in scheduling a software product.
- 9) Define SCM and describe various elements of SCM
- 10) Explain software reviews in detail
- 11) Explain activity network mechanism for concept development project
- 12) What do you mean by defect amplification model? Discuss defect amplification with and without review.
- 13) What is ISO 9000 standard. Define the process of ISO 9000 certification.
- 14) Differentiate between Quality Assurance and Quality Control
- 15) Describe Project manager's objectives.
- 16) Elaborate the concept of Quality Assurance and Quality Control
- 17) Define software reliability and availability.

UNIT – IV

- 1) Explain in detail the effective modular designs.
- 2) Explain the system engineering hierarchy and the factor of system modeling.
- 3) Explain the relationship between modularity and the software cost.
- 4) What do you mean by requirement engineering and what is its guideline principle? Explain.
- 5) Explain the factors which a system engineer should consider while developing a system model.
- 6) Explain the term Quality function development (QFD).
- 7) Explain with the help of neat sketch how an analysis model can be translated into a software design. i.e. the design model.
- 8) Describe the goal of product engineering also draw the product engineering hierarchy.
- 9) Explain the relationship between modularity and the Software Cost.
- 10) What is software prototyping? Explain the throw away and evolutionary prototyping.
- 11) Explain Close Ended and Open Ended prototyping paradigm.
- 12) Discuss the system engineering hierarchy.
- 13) Explain requirement analysis and how it bridges between system engineering and software design?
- 14) Explain design documentation specification. Why do software development projects generate so much documentation?
- 15) Explain the following
 1. Partitioning in detail.
 2. Quality function deployment (QFD) .
 3. Cohesion.
- 16) Explain business process engineering hierarchy?
- 17) Explain Design process in details.
- 18) Write note on design model and documentation.

UNIT – V

- 1) Explain any one of the Golden rule for user interface design
- 2) Explain basic steps in
 - i) Transform Mapping
 - ii) Transaction Mapping
- 3) Explain the steps for component Level Design
- 4) Explain the following terms.
 - i) Tabular design notation
 - ii) Program design notation
- 5) What is architectural style? Explain taxonomy of style and pattern?
- 6) Why high cohesion and low coupling necessary in effective modular design.
- 7) Describe the different evaluation criteria applied when a design model interface has created
- 8) How does the concept of recursive modules fit into the design philosophy and techniques?
- 9) Explain task analysis and modelling
- 10) Explain data design, in detail
- 11) What is software architecture design? State the data design architectural style.
- 12) What is meant by transaction mapping? Discuss various design steps for transaction mapping.
- 13) Explain interface design evaluation cycle in brief.
- 14) Explain data design process and what principles are to be followed in data design.
- 15) Describe in brief the taxonomy of styles and patterns of architecture style.

UNIT – VI

- 1) Explain why regression testing is necessary?
- 2) Explain Following type of testing.
 - i. Smoke Testing
 - ii. White Box Testing
 - iii. Black Box Testing
 - iv. Integration Testing
- 3) Explain the terms.
 - i. Verification
 - ii. Validation
 - iii. Cyclomatic Complexity
 - iv. Debugging
- 4) What is integration testing and explain top down integration and bottom up integration.
- 5) What is smoke testing? Explain in detail.
- 6) Explain white box testing in detail.
- 7) What is debugging? Explain Backtracking and cause Elimination.
- 8) State and explain primary objective for testing.
- 9) Explain any one method of control structure testing.
- 10) Explain basis path testing. Give an example.
- 11) Explain the terms verification and validation. How these help in achieving software quality?
- 12) Explain Cyclomatic complexity.
- 13) What is debugging? What are the bug removal considerations?
- 14) Explain Key quality attributes for computer software.
- 15) Discuss software testing strategy for conventional software architecture, draw suitable diagram.
- 16) Explain any one method of control structure testing.
- 17) Differentiate between verification and validation.